On prior worksheet asked about 2 x 2 matrices which have same column space as null space. Then asked same for 3 x 3.

For 2 x 2 we found some that worked where both spaces were a certain line, i.e. both dimension one.

For 3 x 3 via some cases showed couldn't happen there.

But rank-nullity shows it much easier! Sum of dimensions of column space and null space is three for 3 x 3 matrix. So since that's odd number, the two can't have same value.

Thus column space and null space can't have same dimension so can't possibly be same subspace. So rank-nullity establishes it much more easily!

Good illustration of how important rank-nullity is.

Want to now talk about a concept from 4.4

We are near end of 4.5, I skipped over this earlier since it's useful to have concept of dimension when clarifying how this works.

Consider in $\mathbb{R}^4$ the subspace spanned by the vectors: $\{ (1, 0, 0, 0) , (0, 1, 0, 0) \}$.

Is this subspace $\mathbb{R}^2$? This is a common point of confusion.

In one sense you would say no since the vectors have four components so they aren't even in $\mathbb{R}^2$.

In another sense, geometrically this subspace is the xy-plane so feels like it is $\mathbb{R}^2$.

Hence why this can be confusing as there seems to be a reasonable argument for both sides.

The answer is no, indeed because the vectors aren't even in $\mathbb{R}^2$ so they aren't a subset of $\mathbb{R}^2$ much less a subspace. But surely the fact that geometrically it's equivalent means something?

Point is this subspace of $\mathbb{R}^4$ is **isomorphic** to $\mathbb{R}^2$

For V, W vector spaces (usually real, certainly in this class assume real unless stated otherwise), if T: V ---> W is a one-to-one linear transformation from V onto W, T is an <u>isomorphism</u> and we say V and W are <u>isomorphic</u> to each other.

This effectively means that V and W, even though they might look very different, might be structurally equivalent even if they look completely dissimilar to the naked eye.

For example one space might have its "vectors" as some subspace of some $\mathbb{R}^n$ while the other say has its vectors as a space of functions, e.g. $\mathbb{P}_m$ the space of polynomials of degree $\leq m$ again very quite different looking "vectors", but still it might be that the two spaces are actually still structurally the same, i.e. isomorphic.

Concept of isomorphic is very important in higher level math. Perhaps not used a ton in this class but generally with many structures, vector spaces and otherwise, helpful to gauge if two structures are inherently the same structurally even if they look very different.

Very important result is (one big thing dimension helps with) if V is a real vector space of dimension n, then V is isomorphic to $\mathbb{R}^n$.

How? Take basis for V: $\{v_1, v_2, \dots, v_n\}$ and basis $\{w_1, w_2, \dots, w_n\}$ for $\mathbb{R}^n$ (could be standard basis, doesn't matter) then define $T: V \dashrightarrow \mathbb{R}^n$ via

$$T(v_i) = w_i \qquad \text{for all } 1 \le i \le n$$

as this creates an isomorphism between them. I.e. the w_j's spanning W forces this to be onto, and the v_i's being linearly independent (they are basis like the w_j's) forces this to be one-to-one.

When asking questions about vector space, say were checking for linear independence, often people suggest using matrix ideas to help. But I would immediately say, "wait a second if we aren't in context of $\mathbb{R}^n$, can we really do that?"

As this shows, as long as vector space is finite dimensional, we can! Ok might have to work to set up the isomorphism with the appropriate $\mathbb{R}^n$ but fundamentally it can always be done. An infinite dimensional space is where we might run into trouble with this.

For example, on worksheet gave example of linear transformation via derivative on both $\mathbb{P}$ and $\mathbb{P}_5$.

The former was not representable as matrix transformation since $\mathbb{P}$ is infinite dimensional. The latter was since $\mathbb{P}_5$ is finite dimensional.

In fact for any n we have $\mathbb{P}_n$ isomorphic to $\mathbb{R}^{n+1}$.

How? Take the basis $\{1, x, x^2, \ldots, x^n\}$ for $\mathbb{P}_n$, this is often considered the "standard basis" for $\mathbb{P}_n$

We can say map $T: \mathbb{P}_n \dashrightarrow \mathbb{R}^{n+1}$ by

$$T(x^i) = e_{i+1}$$

for each $0 \leq i \leq n$.

So if we wanted to look at some linear transformation on $\mathbb{P}_n$ into some finite dimensional vector space we could use the matrix transformation:

$$[\,T(1), T(x), T(x^2), \ldots, T(x^n)\,]$$

those being the columns of the matrix, with each expressed as a linear combination of some basis of the codomain. And can use same idea for any linear transformation from a finite dimensional real vector space into another.

4.5.5

$$\left\{ \begin{bmatrix} a - 4b - 2c \\ 2a + 5b - 4c \\ -a + 2c \\ -3a + 7b + 6c \end{bmatrix} : a, b, c \in \mathbb{R} \right\}$$

Subspace? Dimension?

Parametric vector form:

$$a \begin{bmatrix} 1 \\ 2 \\ -1 \\ -3 \end{bmatrix} + b \begin{bmatrix} -4 \\ 5 \\ 0 \\ 7 \end{bmatrix} + c \begin{bmatrix} -2 \\ -4 \\ 2 \\ 6 \end{bmatrix}$$

Span of these so is subspace. And three of them so dimension three? Well careful! Are these linearly independent? Have to check that. They are not since third one is quite clearly multiple of first.

So can say only consider first two for spanning. And since those two are not multiples of each other, they will be linearly independent, guaranteeing dimension is two.

4.5.11

$$A = \begin{bmatrix} 1 & -6 & 9 & 0 & -2 \\ 0 & 1 & 2 & -4 & 5 \\ 0 & 0 & 0 & 5 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Want dimensions of Nul A, Col A, Row A.

Those dimensions all about pivots (already in REF), latter two are exactly number of pivots while first is number of nonpivot columns. Thus blue being pivots,

$$\dim(\text{Nul } A) = 2$$
$$\dim(\text{Col } A) = \dim(\text{Row } A) = 3$$

4.6 Change of Basis

In 4.4 we discussed switching from standard basis of $\mathbb{R}^n$ to some arbitrary basis or vice versa. Now we take it up a notch, learn how to switch between two arbitrary bases.

Honestly not very important for this class. For one this topic is quite isolated, won't really use it later in class. Not like 4.4, 4.5 which get used a great deal later on.

But also practically, the motivation to switch from some arbitrary basis to standard basis is quite clear, the standard basis is much easier to work with usually.

Switching between two arbitrary bases doesn't carry that motivation.

But there are some reasons it could be useful.

In $\mathbb{R}^n$ one might say why work with some other basis than standard basis since standard basis tends to be more convenient?

Well it typically is, but say we had a linear transformation from $\mathbb{R}^n$ to another space, and the associated matrix happened to be very complicated. But in some other basis, the matrix was much more convenient, perhaps a diagonal matrix. That could be motivation to use a different basis (will see this later in course).

Another situation could be, say we had a subspace of some $\mathbb{R}^n$, presumably not the full space, maybe the null, column, or row space of some matrix. Well that subspace won't have a natural attached "standard" basis to it like overall space. So if the basis we get for it via the usual techniques happens to be inconvenient for whatever reason, that could be motivation to switch to something more convenient.

And again since it's not the full space, you don't really have a "standard basis" to switch to.

Take

$$\mathcal{B} = \{b_1, b_2, \dots, b_n\}$$
$$\mathcal{C} = \{c_1, c_2, \dots, c_n\}$$

Suppose these are two bases in $\mathbb{R}^n$ and we want to switch from $\mathcal{B}$ to $\mathcal{C}$. That is, for a vector x if we have $[x]_{\mathcal{B}}$, i.e. the $\mathcal{B}$-coordinates of x, we want to conveniently switch to $[x]_{\mathcal{C}}$, the $\mathcal{C}$-coordinates of x.

We could of course switch from $\mathcal{B}$ to standard basis then to $\mathcal{C}$ but again better if it can be done more conveniently.

How did we switch from $[x]_{\mathcal{B}}$ to x? We did

$$\begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix} [x]_{\mathcal{B}} = x$$

where $b_1, b_2, \ldots, b_n$ are columns, the basis vectors of $\mathcal{B}$ represented in the standard basis (denoted $\mathcal{P}_{\mathcal{B}}$)
That notation was introduced in 4.4, but am only mentioning it now when it really becomes relevant.

Fortunately works same kind of way for switching from $\mathcal{B}$ to $\mathcal{C}$

That is, write the b_i's in terms of the c_j's, and the coefficients of each b_i corresponds to one column of the matrix. That is,

$$
\begin{aligned}
b_1 &= r_{11} c_1 + r_{12} c_2 + \ldots + r_{1n} c_n \\
b_2 &= r_{21} c_1 + r_{22} c_2 + \ldots + r_{2n} c_n \\
&\vdots \\
b_n &= r_{n1} c_1 + r_{n2} c_2 + \ldots + r_{nn} c_n
\end{aligned}
$$

Then the matrix which transforms $\mathcal{B}$ to $\mathcal{C}$ (that is switches a vector's $\mathcal{B}$-coordinates to its $\mathcal{C}$-coordinates) denoted by $\mathcal{P}_{\mathcal{C} \leftarrow \mathcal{B}}$, is given by

$$\begin{bmatrix} r_{11} & r_{21} & \cdots & r_{n1} \\ r_{12} & r_{22} & \cdots & r_{n2} \\ \vdots & \vdots & & \vdots \\ r_{1n} & r_{2n} & \cdots & r_{nn} \end{bmatrix}$$

Why does this work? Take

$$\begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}_{\mathcal{B}}$$

In the standard basis this equals

$$x_1 b_1 + x_2 b_2 + \ldots + x_n b_n$$

Applying $\mathcal{P}_{\mathcal{C} \leftarrow \mathcal{B}}$ to this say gives us

$$\begin{bmatrix} r_{11} & r_{21} & \cdots & r_{n1} \\ r_{12} & r_{22} & \cdots & r_{n2} \\ \vdots & \vdots & & \vdots \\ r_{1n} & r_{2n} & \cdots & r_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$$

Point is the resulting y_j' s are what we believe to be the $\mathcal{C}$-coordinates of this vector, need to verify that. To that end, if they were, then this in the standard basis would equal

$$y_1 c_1 + y_2 c_2 + \ldots + y_n c_n$$

Replacing what the y_j' s actually are, i.e.

$$\begin{aligned} y_1 &= r_{11} x_1 + r_{21} x_2 + \ldots + r_{n1} x_n \\ y_2 &= r_{12} x_1 + r_{22} x_2 + \ldots + r_{n2} x_n \\ &\vdots \\ y_n &= r_{1n} x_1 + r_{2n} x_2 + \ldots + r_{nn} x_n \end{aligned}$$

the total coefficient of x_1 becomes

$$r_{11} c_1 + r_{12} c_2 + \ldots + r_{1n} c_n$$

which is just equal to b_1 from earlier. Thus overall this, doing the others similarly, is:

$$x_1 b_1 + x_2 b_2 + \ldots + x_n b_n$$

which is what we wished to show. I.e. the representation in $\mathcal{B}$ when switched to standard basis is equivalent. to the representation in $\mathcal{C}$ when that is switched to the standard basis. So indeed that matrix switches $\mathcal{B}$ coordinates to $\mathcal{C}$ as we wished to show.

What is best practical way to switch from $\mathcal{B}$ to $\mathcal{C}$?
Two main options:

(1) Write each b_i as linear combination of the c_j's
Can find the coefficients manually in many ways, perhaps variable elimination.

(2) Create augmented matrix:

$$\left[\, c_1 \; c_2 \; \ldots \; c_n \;\middle|\; b_1 \; b_2 \; \ldots \; b_n \,\right] \text{ ----> } n \times 2n$$

convert this to I
via row operations

this will
become $\mathcal{P}_{\mathcal{C} \leftarrow \mathcal{B}}$

Why does this work? Well can view this as solving for the b_i's in terms of the c_j's, several augmented matrices at once essentially.

Or can recall for $[x]_{\mathcal{B}}$ ----> x ----> $[x]_{\mathcal{C}}$

For the first we need to use the b_i's matrix, i.e. $\mathcal{P}_{\mathcal{B}}$
and then for the second, the inverse of the c_j's, i.e. $\mathcal{P}_{\mathcal{C}}^{-1}$

And that's what this does, starts with $\mathcal{P}_{\mathcal{B}}$ on right and $\mathcal{P}_{\mathcal{C}}$ on left. Converting the left half of the augmented matrix to I multiplies the right half by $\mathcal{P}_{\mathcal{C}}^{-1}$ on the appropriate side so overall the right half of the matrix becomes what we need it to be.

That's same thing we did back in learning how to find A^{-1} via augmented matrix $[A \;\; I]$.

4.6.7 $b_1 = \begin{bmatrix} 7 \\ 5 \end{bmatrix}$, $b_2 = \begin{bmatrix} -3 \\ -1 \end{bmatrix}$, $c_1 = \begin{bmatrix} 1 \\ -5 \end{bmatrix}$, $c_2 = \begin{bmatrix} -2 \\ 2 \end{bmatrix}$

Want to find $\mathcal{P}_{\mathcal{C} \leftarrow \mathcal{B}}$

Need to find constants $r_{11}, r_{12}, r_{21}, r_{22}$ so that

$$b_1 = r_{11} c_1 + r_{12} c_2$$
$$b_2 = r_{21} c_1 + r_{22} c_2$$

Inputting the stated vectors in, we then have the two systems:

$$\begin{bmatrix} 7 \\ 5 \end{bmatrix} = \begin{bmatrix} r_{11} - 2r_{12} \\ -5r_{11} + 2r_{12} \end{bmatrix} \quad , \quad \begin{bmatrix} -3 \\ -1 \end{bmatrix} = \begin{bmatrix} r_{21} - 2r_{22} \\ -5r_{21} + 2r_{22} \end{bmatrix}$$

First gives us:

$$7 = r_{11} - 2r_{12}$$
$$5 = -5r_{11} + 2r_{12}$$

Adding these: $12 = -4r_{11}$ ----> $r_{11} = -3$ ----> $r_{12} = -5$

Other one is:

$$-3 = r_{21} - 2r_{22}$$
$$-1 = -5r_{21} + 2r_{22}$$

Adding again: $-4 = -4r_{21}$ ----> $r_{21} = 1$ ----> $r_{22} = 2$

Thus

$$\mathcal{P}_{\mathcal{C} \leftarrow \mathcal{B}} = \begin{bmatrix} -3 & 1 \\ -5 & 2 \end{bmatrix}$$

Other approach:

$$\left[\begin{array}{cc|cc} 1 & -2 & 7 & -3 \\ -5 & 2 & 5 & -1 \end{array}\right] \quad \text{----> Replace R2 with R2 + 5R1}$$

$$\sim \left[\begin{array}{cc|cc} 1 & -2 & 7 & -3 \\ 0 & -8 & 40 & -16 \end{array}\right] \quad \text{----> Replace R2 with } \frac{-1}{8} \text{ R2}$$

$$\sim \left[\begin{array}{cc|cc} 1 & -2 & 7 & -3 \\ 0 & 1 & -5 & 2 \end{array}\right] \quad \text{----> Replace R1 with R1 + 2R2}$$

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -3 & 1 \\ 0 & 1 & -5 & 2 \end{array}\right]$$

same as other method

Again 4.6 not super important, earlier ideas in this chapter much more important. For instance we use ideas of null space often in next chapter.

Be very careful about doing the calculations in this setting, that's the biggest problem I see people having.

That is, you often need to multiply two matrices to create what you need. But naturally you have to be careful about the order as that does matter!

Additionally, sometimes you have to take the inverse of one of the two, that naturally opens the door for mistakes in terms of order / which one you invert.

Also, using augmented matrix, which one is on left? And when setting up matrix, need to not use the transpose of the appropriate matrix.

All those mistakes are common, and I think the reason they happen is people aren't thinking about how change of basis is working, they are drawing on the memory of doing a similar problem and "copying" the nature of the calculation they are remembering.

And that can get you in trouble as there is a decent amount of nuance / different ways problems can be given, you really need to try to understand how it works, not just memorize the formula.